

Raghavendra Prasath Sridhar

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EDUCATION

Northeastern University, Boston, MA

Sep 2024 -- Dec 2026

Master of Science in Information Systems

Coursework: Big Data Systems, Data Science Engineering Methods, Algorithms, Database Design, Application Engineering

Vels Institute of Science, Technology & Advanced Studies, Chennai, India

Aug 2017 -- May 2021

Bachelor of Engineering, Electronics & Communication Engineering

Coursework: Data Structures, Digital Signal Processing, Computer Networks, Digital Image Processing, IoT

TECHNICAL SKILLS

Languages: Python, SQL, R, Java

AI/ML: LLMs, RAG, Agentic Systems, LangGraph, LangChain, MCP, Vector Databases, Prompt Engineering

Data Engineering: Snowflake, PySpark, Airflow, AWS Glue, ETL, Data Modeling

Backend & APIs: FastAPI, REST APIs, Redis, ChromaDB, SQLite

Cloud & DevOps: AWS, Docker, PostgreSQL, Git, Jenkins, GitLab CI/CD, AutoSys, Grafana, Datadog, Prometheus

EXPERIENCE

Fidelity Investments, Boston, MA

Jul 2025 -- Dec 2025

Software Engineer Co-op

- Architected and shipped production automation in Python replacing a **9-hour** end-to-end manual quarterly workflow with real-time generation and a scheduled batch job serving nearly **100 internal users** across portfolio operations
- Designed Python services for workflow orchestration, validation, and exception handling that reduced reviewer rework by **40%** and accelerated sign-off, partnering with business stakeholders to translate ambiguous needs into production specs
- Deployed releases via Jenkins CI/CD with Git-based controls and implemented role-based access control (RBAC) enforcing least-privilege authorization, replacing ad-hoc execution with scheduled, version-controlled, unattended runs
- Instrumented Grafana, Datadog, and Prometheus observability with alerting on application and data health, reducing time-to-detect issues and accelerating incident triage during quarterly cycles
- Built a Snowflake data ingestion and reconciliation layer with AWS integration patterns using window functions, CTEs, self-joins, and UDFs, replacing a multi-day cross-team data pull with auditable certification datasets

Cognizant Technology Solutions, Chennai, India

Dec 2021 -- May 2024

Associate (Data Engineer)

- Owned data migration for **10M+ records** across legacy and target systems, achieving **98%** field-level consistency through SQL reconciliation checks across client production environments
- Cut AWS operational costs **20%** by optimizing S3, Glue, Lambda, EC2, and VPC resource utilization across environments
- Built AWS Glue ETL workflows in Python processing **2TB+** datasets to migrate a legacy Informix/DB2 insurance app to a modern Java architecture
- Reduced query processing time **30%** by redesigning DDL across **50+** PostgreSQL tables and optimizing indexing strategies for analytics workloads
- Deployed AWS RDS schema changes via Liquibase CI/CD on GitLab, replacing manual SQL with version-controlled, rollback-safe releases

PROJECTS

Portfolio Intelligence System --- Agentic AI Due Diligence Platform

Jan -- May 2026

LangGraph, FastMCP, FastAPI, Snowflake, ChromaDB, Airflow, Redis

- Built the data-engineering backbone of a multi-phase agentic AI platform automating private-equity due diligence across **10 portfolio companies**, spanning SEC EDGAR ingestion, a normalized Snowflake schema, scoring persistence, and a full audit trail
- Co-built hybrid retrieval (BM25 + sentence-transformer embeddings + RRF with HyDE) over SEC filings and integrated a LangGraph supervisor with an MCP tool server, validating reliability with **157** unit, integration, and property-based tests

ReviewPulse AI --- Cross-Platform Review Intelligence

Mar -- Apr 2026

PySpark, Snowflake, DistilBERT, ChromaDB, Python

- Orchestrated a distributed PySpark pipeline ingesting **580M+ reviews** across five platforms (Amazon, eBay, Yelp, iFixit, YouTube), producing **8.1M+** normalized, sentiment-scored rows in a Snowflake warehouse
- Engineered DistilBERT sentiment scoring with sentence-transformer embeddings indexed in ChromaDB for semantic search, and delivered a Streamlit dashboard with a human-in-the-loop review queue

Block Explorer AI --- LLM-Powered Bitcoin Text-to-SQL Explorer

May -- Jun 2026

Bitcoin Core, Bitcoin RPC, Python, SQLite, OpenRouter LLMs, Streamlit

- Built an end-to-end pipeline letting users query live Bitcoin data in plain English, syncing a Bitcoin Core node of **110K+ blocks** and **289K+ transactions** via RPC into a normalized SQLite database with deterministic ingestion and validation
- Engineered a text-to-SQL engine injecting live schema into an LLM with a read-only SELECT-only safety guard and a CANNOT_ANSWER guardrail, validated by **12/12** golden test cases plus documented LLM hard-failure analysis

PUBLICATIONS & LEADERSHIP

- Published peer-reviewed ResNet-50 brain tumor classification research achieving **95.3%** accuracy in AIP Conference Proceedings, later cited by Stanford researchers (doi.org/10.1063/5.0082328)